

**K S Tharunpranav 251801359-2025-AIDS** ▾**K2****Started on** Thursday, 17 September 2026, 8:10 AM**State** Finished**Completed on** Thursday, 17 September 2026, 8:15 AM**Time taken** 4 mins 31 secs**Marks** 1.00/1.00**Grade** **10.00** out of 10.00 (**100%**)

Question 1 | Correct | Mark 1.00 out of 1.00

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

s1		a		g		g		t		a		b
s2		g		x		t		x		a		y
										b		

The length is 4

Solveing it using Dynamic Programming

For example:

Input	Result
aab	2
azb	

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  #include <string.h>
3  int max(int a, int b) {
4      return (a > b) ? a : b;
5  }
6  int main() {
7      char s1[1005];
8      char s2[1005];
9      if (scanf("%s", s1) == 1 && scanf("%s", s2) == 1) {
10         int m = strlen(s1);
11         int n = strlen(s2);
12         int dp[m + 1][n + 1];
13         for (int i = 0; i <= m; i++) dp[i][0] = 0;
14         for (int j = 0; j <= n; j++) dp[0][j] = 0;
15         for (int i = 1; i <= m; i++) {
16             for (int j = 1; j <= n; j++) {
17                 if (s1[i - 1] == s2[j - 1]) {
18                     dp[i][j] = dp[i - 1][j - 1] + 1;
19                 } else {
20                     dp[i][j] = max(dp[i - 1][j], dp[i][j - 1]);
21                 }
22             }
23         }
24         printf("%d\n", dp[m][n]);
25     }
26     return 0;
27 }
28

```

	Input	Expected	Got	
✓	aab azb	2	2	✓
✓	ABCD ABCD	4	4	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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